

Chapter – 3: Health: The Ultimate Treasure

Health: Is It More Than Not Falling Sick?

- Being healthy – not just disease-free life – BUT also – good physique, positive attitude, strong rel.
- Healthy person – maintain – body, positive mindset, social life
- **Explore** – true meaning – health
- Activity –
 - Grade 8 std. – moves – new school, new city – no friends – new env. – busy parents – very lonely
 - To cope – increased screen time, social media – feel more worse
 - Stopped trying – make friends – headaches – lost weight – no sleep
 - Doctor advise – less screen time – meet counsellor – school counsellor – arrange help – support him – make friends – improve health
 - **Think, reflect** – cause – boy's health problems? – habits, surroundings – affect – well-being
- World Health Organisation (WHO) – defines health – state – phy, mental, social well-being – NOT only – absence – disease
- Healthy person – various tasks – efficiently – cope well – diff., difficult situations – also – adjust well – pee grps, other members – society

How can we stay healthy?

- to stay healthy – eat nutritious food – maintain hygiene – stay – clean place – regular exercise – proper sleep – spend time – friends, family – positive attitude
- What to do? What to avoid?
- Activity –
 - List – good habits – parents, teachers, elders – recommend – how many – already part – routine? – which ones – start following? – add – below list –
 - Keep clean – maintain hygiene
 - Eat healthy, balanced diet
 - Regular exercise
 - Make time – relax, meditate – daily
 - Think – habits – not good – health – add – below list –
 - Spend time – mobile phones – digital screens
 - Eat – fast food, other junk – daily
 - Sleep late – not enough sleep
 - Skip meals – especially breakfast
- Take care – body, mind – imp. – healthy habits – support healthy body – healthy mind
- Discuss findings – friends, teachers
- Act., discussions – realise – health – depend – many factors – incl. – lifestyle, env.

Maintain healthy lifestyle

- Eat – balanced diet – plenty – fruits, vegetables, whole grains
- Avoid – processed, fatty, sugary – food, drinks
- Stay active – play outdoors, walk, run, cycle, exercise
- Limit – screen time – more time – nature

- Enough sleep – help body, mind – rest, recover
- Practice – yoga, breathing exercise – pranayama
- Say NO – harmful substance – tobacco, alcohol, drugs

Keep the environment clean

- Activity –
 - Look fig. – which playground – play? – why?
 - Most – play – 1st playground – clean, well-maintained, beautiful – 2nd playground – polluted, dirty, unhygienic, flies, mosquitos – people – such areas – fall sick often
- Addition – inculcating good habits – healthy lifestyle – ourselves, surroundings clean
- Ever found – breathing hard – smoke, dust – clean air, water – imp. – health – cities – air pollution – vehicles, factories – problems – coughing, asthma – **Air Quality Index (AQI)** – help know – how clean – air? – cleaner env. – help stay – healthy, feel better
- Health – NOT only body – BUT also – feelings, rel. – matter as well
- Even – eat well – live – clean place – not feel good – lonely, upset
- Spend time – friends, family – talk, laugh, have fun – keep mind – healthy

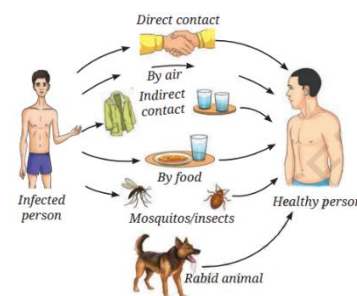


How Do We Know That We Are Unwell?

- Our body – works – certain way – keep healthy – feel unwell – means – something inside – not working
- Have symptoms – pain, tiredness, dizziness – signs – fever, rash, high B.P., swelling – indicate – unwell
- Symptom – we feel – sign – something – seen, measured – help doctors – understand – what went wrong

Diseases: What Are the Causes and Types?

- Disease – condition – affect – normal working – body, mind – happens – 1 or more – organ, organ sys. – stop proper func.
- Some diseases – germs – bacteria, virus, fungi, worms, protozoa – **pathogens**
- Other diseases – result – poor nutrition, unhealthy lifestyle
- Some diseases – short time – others – long time – regular treatment, care
- Diseases – grp. – 2 major types – basis – causes, spread –
 - **Non-communicable diseases** – some diseases – cancer, diabetes, asthma – no pathogens – no spread – linked – lifestyle, diet, env.
 - **Communicable diseases** – disease – by pathogens – spread – person to person – ex. – typhoid, dengue, flu, chickenpox, COVID-19
- Recent yrs. – non-communicable diseases (NCDs) – diabetes, heart disease, cancer – more common – India
- Reason – changes – lives – more processed food, less exercise, longer lives
- Most deaths – India – NCDs – understand diff. – 2 types – help know – disease spread – prevent them



How are communicable diseases caused and spread?

- Communicable diseases – cause – pathogens – enter body – air, contaminated food, water

- BUT – how – pathogens – spread – person to person?
- Common way – air – infected person – cough, sneeze – OR – direct contact – shake hands – OR – indirectly – share personal items – infected person
- Some communicable diseases – spread – contaminated food, water
- Some pathogens – spread – insects – mosquitos, houseflies – called **vectors**
- Understand – disease spread – simple steps – protect ourselves, others
- Activity –
 - Grade 8 std. – list – common communicable diseases – community campaign – library search
 - Check listed info. – refer books, internet, teachers – add – missing details
 - Study table – think – simple steps – prevent diseases

Diseases	Causal agent	Site of infection	Symptoms	Preventive measures
Disease spread through air				
Common cold and influenza	Virus	Respiratory tract	Nasal congestion, discharge, soar throat, cough, body ache	Wash hands – no sharing – personal items – using mask
Chicken pox	Virus	Respiratory tract, skin	Mild fever, itchy skin, rashes, blisters	Complete isolation – patient – use mask – get vaccinated
Measles	Virus	Skin, respiratory tract	Fever, soar throat, reddish rashes – neck, ears, other parts – skin	Isolation – patient – cover mouth, nose – maintain – hygiene, vaccination
Tuberculosis	Bacteria	Lungs	Cough, fever, fatigue, loss of appetite, night sweats	Avoid close contact – infected people – cover mouth, nose – maintain hygiene – get vaccination
Diseases spread through contaminated water, food				
Hepatitis A	Virus	Liver	Fatigue, fever, loss of appetite, nausea, vomiting, jaundice, pain – upper right abdomen	Drink boiled water, vaccination
Cholera	Bacteria	Intestine	Diarrhea, dehydration	Maintain – personal hygiene, sanitary habits, consume – proper food, boiled water – vaccination
Typhoid	Bacteria	Intestine	Headache, abdominal discomfort, fever, diarrhea	Maintain – personal hygiene, sanitary habits, consume – proper food, boiled water – vaccination
Ascariasis (roundworms)	Worms	Intestine	Worms in stool, loss of appetite, poor growth, wight loss, anaemia	Maintain – personal hygiene, sanitary habits, consume –

				proper food, boiled water
Diseases transmitted by insects				
Malaria	Protozoa	Skin, blood	High fever, profuse sweating, periodic chills	Use – mosquito nets, repellents – long-sleeve clothes – control – mosquito breeding – home
Dengue fever	Virus	Skin, blood	Fever, headache, muscle, joint pain, nausea	Use – mosquito nets, repellents – long-sleeve clothes – control – mosquito breeding – home – avoid – still waters

- Study table – understand spread – infectious diseases – prevent them
- Some simple – BUT – imp. precautions –
 - Keep – ourselves, surroundings – clean
 - Practice basic hygiene – daily
 - Wash hands – soap, water – remove pathogens
 - Cover – mouth, nose – coughing, sneezing
 - Wear mask – crowd place – protection
 - Avoid sharing – personal items – towels, handkerchiefs
 - Stay home – rest – unwell – help body – recover – minimize spread
 - Keep – home, food, water – clean
- Some disease – cause – worms – inside body – especially digestive sys. – feed – nutrients – **parasites**
- These worms – spread – contaminated food, water, soil, contact – infected person, animals

How are non-communicable diseases caused?

- Learnt – non-communicable diseases – cancer, diabetes, asthma – link – lifestyle, diet, env.
- India – death – most common cause – these diseases
- Grade 6 – also learnt – diseases – scurvy, anaemia, goitre – cause – lack – specific nutrients – diet – **deficiency diseases** – also non-communicable
- Diseases – cancer, diabetes, asthma – persist – long time – referred – **chronic diseases**
- Diabetes – common diseases – prevalent – adults, children
- India – now – highest no. – diabetic people
- Develop – combo. – hormonal imbalance – unhealthy eating habits – lack – phy. act. – overweight – obesity – etc.
- Activity –
 - Find – 3 most common lifestyle diseases – neighbourhood
 - Talk – doctor, nurse, health worker, family members – knows – health – lifestyle changes – prevent, manage diseases
 - Also consult – health websites, teachers – make table – learn more – lifestyle diseases

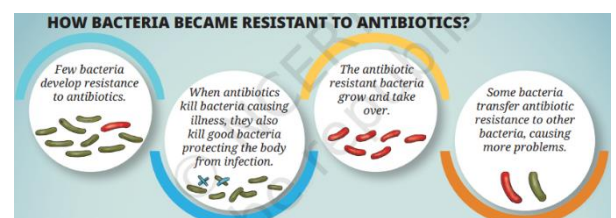
How to Prevent and Control Diseases?

- Heard phrase – prevention is better than cure

- Imp. – protect – ourselves – both – communicable, non-communicable diseases
- **Odisha – community led sanitation program**
 - Bhadrak district – Odisha – comm. sanitation campaign – help people – build, use toilets
 - Open defecation – reduced fast – improve child health – fewer cases – diarrhea, infections
- **Infer** – this case – simple steps – good sanitation – reduce spread – communicable diseases
- Find – comm. campaigns – your locality – share, discuss – peers – impact – such initiatives
- **Ability of the body to fight diseases**
 - Notice – some people – sick more – living – same env.
 - Natural ability – our body – fight disease – **immunity**
 - Body – special sys. – immune sys. – fight diseases
- Childhood – drops, injections – protection – certain diseases – polio, measles, tetanus, hepatitis
- These – **vaccines** – prevent serious infections – viruses, bacteria
- Vaccine – help body – fight diseases – train immune sys. – recognise, attack harmful germs – provide – **acquired immunity** – protection after exposure – pathogen, vaccine
- Vaccines – made – diff. ways – weakened, dead pathogens – OR – inactive, harmless parts – pathogen
- Newer vaccines – instruct body – make harmless parts – germ – immune sys. – learns to fight
- Ex. – tetanus shot – after injury – protection – infection – tetanus bacteria
- Contains – inactivated bacterial toxin – helps – immune sys. – protection – without disease
- **Edward Jenner and the smallpox vaccine**
 - Smallpox – deadly disease – blisters – killed millions
 - Milder disease – cowpox – cows – also infect humans
 - Late 1700s – English doctor Edward Jenner – discover – people with cowpox – no smallpox
 - Led – invention – 1st vaccine – help protect people – smallpox
- Vaccines – most effective way – protect people – all ages – infants to elderly – many serious diseases
- Prevent illness – reduce spread – infections – save million lives – each yr.
- Imp. – remember – vaccines – prevent – NOT cure – minimize serious disease before happening – BUT – don not treat – someone sick
- Some people – fear, doubt vaccines – BUT – scientists, doctors – careful tests – safety
- Vaccination – protect you – BUT also – people around you

Treatment of diseases

- Immune sys. – fails – protection – infectious disease – fall ill – visit doctor
- Doctor – medicines – **antibiotics** – kill bacteria – cause disease
- Antibiotics – only against – bacterial infections – target – parts – bacterial cells – diff. – humans, other animal cells
- Do not work – diseases – viruses, protozoa
- Antibiotics – effective – protection – bacterial infections – saved million lives – BUT – indiscriminate use – decline – effectiveness
- Nowadays – news headlines – **antibiotic resistance** – phenomenon – bacteria – earlier killed – found – survive, multiply – even after treatment
- Makes – common infections – harder treatment – increase risk – complications, prolonged illness, even death
- Activity –
 - Study – infographic – given fig. – how – antibiotic resistance – developed – bacterial pathogens? – what precautions – taken – reduce resistance?



- Tackle problem – antibiotic resistance – must use – antibiotics wisely – only – prescription – doctor – correct dose – right duration
 - Avoid – unnecessary use – help prevent – rise – resistant bacteria – keep antibiotic effect – future gen.
- Traditional medicine sys. – Ayurveda, Siddha, Unani – used – India – many yrs. – common health problems
- Use natural substance – herbs, oils, minerals – manage illness – promote recovery – focus – healthy lifestyle, balanced diet
- These sys. – help – some conditions – useful – everyday well-being – BUT – may not be effective – all diseases – all stages
- Therapies – non-communicable diseases – focus – manage symptoms – improve quality – life – medication, lifestyle changes, rehabilitation
- Early diagnosis – continuous care – key – control – disease progression – prevent complications